

University of Glasgow

Friday 9 May 2026

14:00 – 15:00 BST

Duration: 1 hour | CLOSED BOOK

DEGREE OF MSc

## Deep Learning for MSc COMPSCI5103

Answer BOTH Questions. Total: 40 marks. CLOSED BOOK — Only the attached Crib Sheet is permitted.

### CRIB SHEET (Provided — refer as needed)

Sigmoid:  $a(z) = 1/(1+\exp(-z))$   $da/dz = a(1-a)$

Tanh:  $a(z) = (\exp(z)-\exp(-z))/(\exp(z)+\exp(-z))$   $da/dz = (1-a^2)$

ReLU:  $a(z) = \max(0, z)$   $da/dz = 1$  if  $z \geq 0$ , else 0

Conv/Pool output size:  $W_{out} = \text{floor}((W_{in} - F + 2P) / S) + 1$

### Question 1 [20 marks]

You should answer the following questions by examining the PyTorch model below.

```
class ConvNet(nn.Module):
    def __init__(self):
        super(ConvNet, self).__init__()
        self.conv1 = nn.Conv2d(1, 8, kernel_size=3, stride=1, padding=1)
        self.conv2 = nn.Conv2d(8, 16, kernel_size=3, stride=2, padding=1)
        self.fc1 = nn.Linear(16 * 8 * 8, 64)
        self.fc2 = nn.Linear(64, 3)

    def forward(self, x):
        x = F.relu(self.conv1(x))
        x = F.relu(self.conv2(x))
        x = x.view(x.size(0), -1)
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return x

criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001, weight_decay=0.01)
```

(a)

The input to this network is a tensor of shape **torch.size([4, 1, 32, 32])**. Describe what real-world data this could represent and explain what each of the four dimensions means.

[3 marks]

**(b)**

Calculate the total number of parameters in this network. Show your working clearly for each layer. State the formula you are using.

**[5 marks]**

**(c)**

Trace the tensor shape through each layer of the *forward* method when the input shape is **[4, 1, 32, 32]**. Write the output shape next to each layer name.

**[4 marks]**

**(d)**

The final layer **fc2** outputs raw values with no activation function, and **nn.CrossEntropyLoss** is used.

- (i) Is this correct? Explain why or why not, referencing what CrossEntropyLoss does internally. [2]
- (ii) Describe the shape and datatype of the ground truth target tensor required by CrossEntropyLoss for this model. [1]

**[3 marks total]**

**(e)**

The optimizer uses **weight\_decay=0.01**.

- (i) What does weight\_decay do mathematically? Write the modified loss function it produces. [2]
- (ii) What problem does it help solve, and what would you observe in the training/validation loss curves if this problem were present without weight\_decay? [2]

**[4 marks total]**

**(f)**

Adam is used instead of plain SGD. Name and explain **two key algorithmic advantages** Adam has over plain SGD, and describe what type of loss surface each advantage helps with.

**[4 marks]**

## Question 2 [20 marks]

(a)

The following **3×3 kernel** is applied to the **5×5 grayscale image** shown below.

Kernel:

|    |    |    |
|----|----|----|
| 0  | -1 | 0  |
| -1 | 4  | -1 |
| 0  | -1 | 0  |

Image:

|   |    |    |    |   |
|---|----|----|----|---|
| 0 | 0  | 0  | 0  | 0 |
| 0 | 10 | 10 | 10 | 0 |
| 0 | 10 | 20 | 10 | 0 |
| 0 | 10 | 10 | 10 | 0 |
| 0 | 0  | 0  | 0  | 0 |

(i) What type of image feature does this kernel detect? [1]

(ii) Calculate the full output tensor with **padding=0, stride=1**. Show working for at least three positions. [4]

[5 marks total]

(b)

A **MaxPool2d(2×2, stride=2)** layer is applied to the output of part (a)(ii).

(i) What is the output size? Show your working using the formula. [1]

(ii) Write down the MaxPool output values. [2]

[3 marks total]

(c)

A vanilla RNN uses scalar values with the following formulae:

$$\begin{aligned}h_t &= \text{ReLU}(w * h_{(t-1)} + u * x_t) \\y_t &= c + v * h_t\end{aligned}$$

Assume **u = 2.0, v = -1.0, w = 0.5, c = 3.0** and **h<sub>-1</sub> = 0.0**. Two recurrent units are joined together.

Do a **forward pass** using input sequence **x<sub>0</sub> = 2.0, x<sub>1</sub> = -3.0**. Calculate **h<sub>0</sub>, y<sub>0</sub>, h<sub>1</sub>, y<sub>1</sub>**. Show all working clearly.

[5 marks]

(d)

Using the same RNN from part (c) and its general variables (u, v, w, c), determine the expression for **dy<sub>0</sub>/dx<sub>0</sub>**. Consider both cases: when the ReLU input is **≤ 0** and when it is **> 0**.

[4 marks]

(e)

A Transformer self-attention module has the following configuration: **1 attention head, embedding dimension  $d_e = 64$ , key/query dimension  $d_k = 16$ , value dimension  $d_v = 32$ , output dimension  $d_o = 64$ .**

(i) Write the self-attention formula:  $\text{Attention}(Q, K, V) = ?$  Explain what Q, K, V represent in one sentence each. [2]

(ii) Calculate the total number of parameters in this single-head self-attention module (including the output projection). Show your working. [3]

**[5 marks total — (i) 2 marks, (ii) 3 marks]**

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**END OF QUESTION PAPER**